



Consequences of Thailand's 2006 military coup: Evidence from the synthetic control method[☆]

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ABSTRACT

In this paper, I study the consequences of Thailand's 2006 military coup for various dimensions of Thailand's economy. I use the synthetic control method to construct synthetic Thailand, which represents how Thailand would have been if the 2006 coup had not occurred. There are three main results. First, the synthetic control method does not detect statistically significant changes in real GDP, GDP growth rate, inflation, investment, government spending, exports, imports, labor force participation, unemployment, or financial stability. Second, the 2006 military coup temporarily decreased consumption as a percentage of GDP. Third, the military coup increased military expenditure and international tourism in the short run.

1. Introduction

As a constitutional monarchy, Thailand's politics is unique in that the monarchy is actively involved in the process through the Thai King's proxies. This phenomenon is termed a “network monarchy”, which indicates that the monarchy acquired political power and wealth through its network (McCargo, 2005). In the early 2000s, the monarchy's network was under threat from a new prime minister, Thaksin Shinawatra, who had gained popularity in rural areas and had been elected in 2001. Thaksin Shinawatra intended to secure an absolute majority of votes and construct a one-party government, which would have given him and his network monopolistic power (Kongkirati, 2014).

One starting point was the competition between the Shinawatra business group and the monarchy's Crown Property Bureau (Hewison, 2008; Ouyyanont, 2008). The monarchy's fear of losing influence sparked a massive political campaign against Thaksin Shinawatra and his network. Strategic actions between the monarchy and Thaksin Shinawatra's network resulted in political tensions in 2005–2006. The group of people called “People's Alliance for Democracy” or the “Yellow Shirts” protested against Thaksin Shinawatra and put pressure on the military to stage a coup and remove him from power.

Around the same time, tensions between Thaksin Shinawatra and the military began to rise after Thaksin Shinawatra's government cut the budget for military expenditure (Pathmanand, 2008). Subsequently, the monarchy and the military staged a coup to remove Thaksin from power and protect the elites' interests (Chachavalpongpon, 2009). Existing theories such as Acemoglu et al. (2019) and Naidu et al. (2021) suggest that reversals occur when democracy threatens the interests of elites who have the power to overthrow it. The 2006 coup was the monarchists' last effort to take control of the democratic process, notwithstanding the monarchy's apparent “above politics” status (Winichakul, 2008).

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The goal of the 2006 military coup was to eliminate this threat and maintain the power and popularity of the monarchy while preventing future resistance by the public. Therefore, the leaders of the military coup portrayed themselves as saviors who had resolved political tensions and minimized the negative effects on the citizens (Chachavalpongpon, 2009). In contrast to military coups in other countries' histories, Thailand's 2006 military coup was unable to flagrantly abuse its power in a way that would encourage additional political turmoil. This suggests that neither the economy nor the people's quality of life should have deteriorated.

This paper aims to investigate the magnitudes of the effects of Thailand's 2006 military coup on Thailand's economic outcomes: (i) GDP and inflation, (ii) the components of GDP, (iii) the labor market, (iv) financial stability, and (v) military expenditure. Furthermore, several countries such as Canada, the United States, Japan, South Korea, Malaysia, the United Kingdom, and European countries advised their citizens not to travel to Thailand due to safety concerns, released statements against Thailand's military government, and threatened to cut financial aid to Thailand. This paper will also study the effects on international tourism and the United States' bilateral aid flows.

The analysis in this paper uses the synthetic control method, introduced by Abadie and Gardeazabal (2003) and used in Abadie et al. (2010, 2011, 2015), to construct synthetic Thailand as a counterfactual. The synthetic control method has gained popularity recently, having been employed in a number of studies, including Alrababa'H et al. (2021), Gilens et al. (2021), Silva (2018), and Riera (2020). Abadie (2021) provides an overview of recent advances in the synthetic control method. Synthetic Thailand is constructed from a linear combination of countries in the control group, such that synthetic Thailand behaves as if it were actual Thailand in the period before the coup. The behavior of synthetic Thailand after the coup serves as a counterfactual had the 2006 coup not occurred. To estimate the effect on economic variables resulting from the 2006 coup, I compare the post-coup economic variables of actual Thailand and synthetic Thailand.

The main analysis uses data from 2001–2014, five years before the coup and eight years after. The five-year pre-coup period begins after Thailand recovered from the 1997 financial crisis, and the period matches the time when Thaksin Shinawatra was prime minister. The control group consists of the countries in Southeast Asia and the major trading partners of Thailand. For the robustness of the results, I conduct two additional analyses in which the length of the pre-coup period is extended to 10 years, from 1996 to 2006, so that synthetic Thailand can capture the pre-coup long-term trend. The first additional analysis uses the same set of countries in the control group. The second additional analysis includes upper-middle-income countries so that similar upper-middle-income countries can capture the patterns of economic growth and developments in Thailand.

The synthetic control method does not detect statistically significant changes in real GDP, GDP growth rate, inflation, investment, government spending, exports, imports, labor force participation, unemployment, or financial stability. The synthetic control method suggests that consumption as percentage of GDP decreased. The gap of consumption as a percentage of GDP between actual Thailand and synthetic Thailand is around -1.5 and -3.0 percentage points of GDP between 2006 and 2008 before it becomes around -4 percentage points of GDP between 2009 and 2011. The effect on government expenditure as a percentage of GDP is positive after 2009; however, this effect could be due to the financial crisis in 2008.

The empirical analysis detects economically and statistically significant increases in military expenditure and international tourism. Under Thaksin Shinawatra's government, the budget for military expenditure decreased from 5.6 percent of government expenditure in 2001 to 2.2 percent in 2005. The synthetic control method predicts that the actual military expenditure in 2008–2014 was 53–71 percent more than it would have been had the military coup not occurred. The difference is equal to 0.28–0.66 percent of GDP and is statistically different from zero. In 2006 and 2007, actual Thailand had 15.6 and 6.0 percent more international tourists than synthetic Thailand did. The difference corresponds to a 26.5–33.4 percent increase in income from international tourists. The synthetic control method finds a sizable positive effect on international tourism from 2009 to 2014, but the effect is not statistically significant.

This paper is closely related to three branches of the literature. First, it contributes to the literature on the consequences of coups. Kawaura (2018) finds that after each of the nine coups that occurred in Thailand between 1949 and 2014, military expenditure in the following year increases by approximately 23 percent, while this work finds that military expenditure increases by 38 percent in one year and around 60 percent in each of the following five years. Zuk and Thompson (1982) find mixed results on military expenditure. Kim et al. (2013) revisit Zuk and Thompson (1982) with a more extended time period and find a positive effect of a military coup on military expenditure. Leon (2014) provides empirical evidence that successful coups raise military expenditure more than unsuccessful attempts, and coups often happen after military expenditure as a percentage of GDP is low. Bove and Nisticò (2014) present evidence that successful coups result in a large increase in the military burden. According to Akkemik et al. (2020), the failed coup attempt in Turkey in 2016 had a substantial negative effect on people's well-being, happiness, life satisfaction, and trust. Tusalem (2014) argues that in countries where the military is heavily involved in politics, democratic development is more difficult. In contrast, Thyne and Powell (2016) document that coups encourage democracy because leaders are incentivized to democratize rapidly in order to create political legitimacy and economic development. The 2006 military coup studied in this paper is different from other coups in the world's history because it was peaceful and received support from a large number of Thai people. The rationale behind the coup was complex and political. Therefore, many dimensions of the economic outcome of the coup differed from the conclusions drawn by previous studies in the literature. Second, in terms of methodology, this paper is related to a growing body of work that uses the synthetic control approach to investigate the impacts of political changes. Examples include Grier and Maynard (2016), who examine the economic implications of Hugo Chavez's regime, Suwanprasert (2023), who studies Thailand's 2014 military coup, and Uhr et al. (2017), who examine the economic effect of Chile's economic reforms and open democracy. This paper uses the synthetic control method to investigate the economic aftermath of Thailand's 2006 military coup.

The third contribution is to the literature that studies the impacts of Thailand's 2006 military coup. Previous works often focus on the financial markets. For example, [Khanthavit \(2019\)](#) studies the effect on the Thai bond market; [Civilize et al. \(2015\)](#) study the effect on stock returns; and [Rujirarangsarn and Chancharat \(2019\)](#) study the long-run relationship between the SET index and the exchange rate of the Thai baht and the US dollar. Other previous works consider political uncertainty around the coup. For example, [Kongprajya \(2010\)](#) examines how political news in the period 2004–2008 affected stock prices. [Luangaram and Sethapramote \(2018\)](#) find that political uncertainty negatively affects the Thai economy, especially private investment, in the short run and economic growth in the long run. This paper examines a more comprehensive set of outcome variables in a longer time period to get a bigger picture of the economy. For example, this paper studies the effects on tourism, military spending, and aid flows from the United States.

The remainder of the paper is structured as follows. Section 2 provides historical background on Thailand's 2006 coup. Section 3 offers arguments for why the effects of the 2006 coup may not be obvious. Section 4 describes the synthetic control method and the data used in this study. Section 5 discusses the empirical findings from the synthetic control method. Section 6 concludes.

2. Historical background on Thailand's 2006 coup

This section provides historical background on Thailand's politics from 1997 to 2014. A timeline for the key events in this period is provided in [Fig. 1](#).

On September 19, 2006, the Royal Thai Army staged a coup against the elected government of Prime Minister Thaksin Shinawatra. On September 21, the new authorities, led by General Sonthi Boonyaratglin, issued a statement outlining the reasons for seizing power and promising to restore democracy within a year. They then nominated General Surayud Chulanont as interim prime minister. On December 23, 2007, elections were conducted after Thaksin Shinawatra's Thai Rak Thai (TRT) party was banned.

Thaksin Shinawatra is a Thai entrepreneur who established the TRT political party in 1998. He was elected Prime Minister in 2001 after a historic election triumph, and he is the first Prime Minister in Thailand's history to spend a full term in the position. Thaksin Shinawatra's economic efforts helped Thailand's recovery from the 1997 Asian financial crisis, with GDP increasing from 4.9 trillion baht in 2001 to 7.1 trillion baht in 2006.

Thaksin Shinawatra introduced two major health policies: subsidized universal health care at the cost of 30 baht per visit (approximately \$1) and low-cost universal access to anti-retroviral HIV medicines (ARVs). The 30 baht program, which helped raise the rate of access to health care from 76 percent to 96 percent, garnered widespread popular support. Physicians and health care professionals, on the other hand, have criticized the program for producing heavy workloads, overcrowded waiting rooms, and insufficient time to diagnose each patient.

Thaksin Shinawatra legalized the underground lottery numbers game and used the revenue to fund social projects such as the "One District, One Scholarship" project. He also implemented a debt-restructuring plan, a 1-million-baht (approximately \$30,000) community fund, and several government-funded infrastructure projects. His foreign policy intended to promote his populist policies and expand his businesses ([Chachavalpongpun, 2009](#)). The TRT party won the 2005 general election, which had the greatest voter participation in Thai history.

In early 2006, Thaksin Shinawatra was accused of policy corruption that resulted in 5–30 percent more spending by the government than was required. Furthermore, Thailand's Board of Investment (BOI) gave Shin Satellite 16.4 billion baht in tax incentives in 2003, and the Transport Ministry abolished the minimum plane ticket requirement when Shin Corporation finalized a joint venture with AirAsia. The Shinawatra family was accused of endangering national security after selling their Shin Corporation shares to Temasek Holdings, a Singaporean holding company. A group of middle-class people protested against Thaksin Shinawatra on the grounds that he was abusing the middle class for his own personal gain and evading paying taxes on the sale of the Shin Corporation ([Hewison, 2010](#)).

Beginning in February 2006, hundreds of thousands of people protested as part of an organization known as the People's Alliance for Democracy (PAD), or the "Yellow Shirts". On February 24, 2006, Thaksin Shinawatra declared the dissolution of parliament, and a general election was set for April 2. Despite widespread opposition, Thaksin Shinawatra's TRT Party gained 462 seats in parliament; nevertheless, the elections were declared invalid by the Constitutional Court on May 8, 2006. A new election was planned for October 15, 2006.

On September 19, 2006, while Thaksin Shinawatra was attending a United Nations summit in New York City, the army seized control of Bangkok. Tanks and military vehicles were stationed outside Government House, while troops surrounded the Thaicom satellite receiving station and the government-owned television station. During the bloodless coup, anti-Thaksin protesters brought flowers and food to soldiers on duty, which was a surprising occurrence ([Prasirtsuk, 2007](#)).

The military, known as the Council for Democratic Reform under the Constitutional Monarchy (CDRM), published a statement claiming the government's *lèse majesté*, corruption, interference with state agencies, and creation of social divisions as reasons for the coup. General Sonthi Boonyaratglin, the commander of the coup, claimed that national security might be endangered by Temasek's takeover of the Thai telecommunications company Shin Corporation. In addition, Thaksin Shinawatra was accused of corruption in a soft-loan package offered to Myanmar in early 2004, which resulted in the purchase of Shin satellite services for 600 million baht. The CDRM announced that it would restore a democratic government within one year and appointed retired General Surayud Chulanont as interim prime minister.

After the coup, another group of people, called The United Front for Democracy Against Dictatorship (UDD) or the "Red Shirts", started to protest against the military government. On October 2, 2006, Thaksin Shinawatra resigned from the TRT party, and on May 30, 2007, the Thai Supreme Court disbanded the TRT Party.

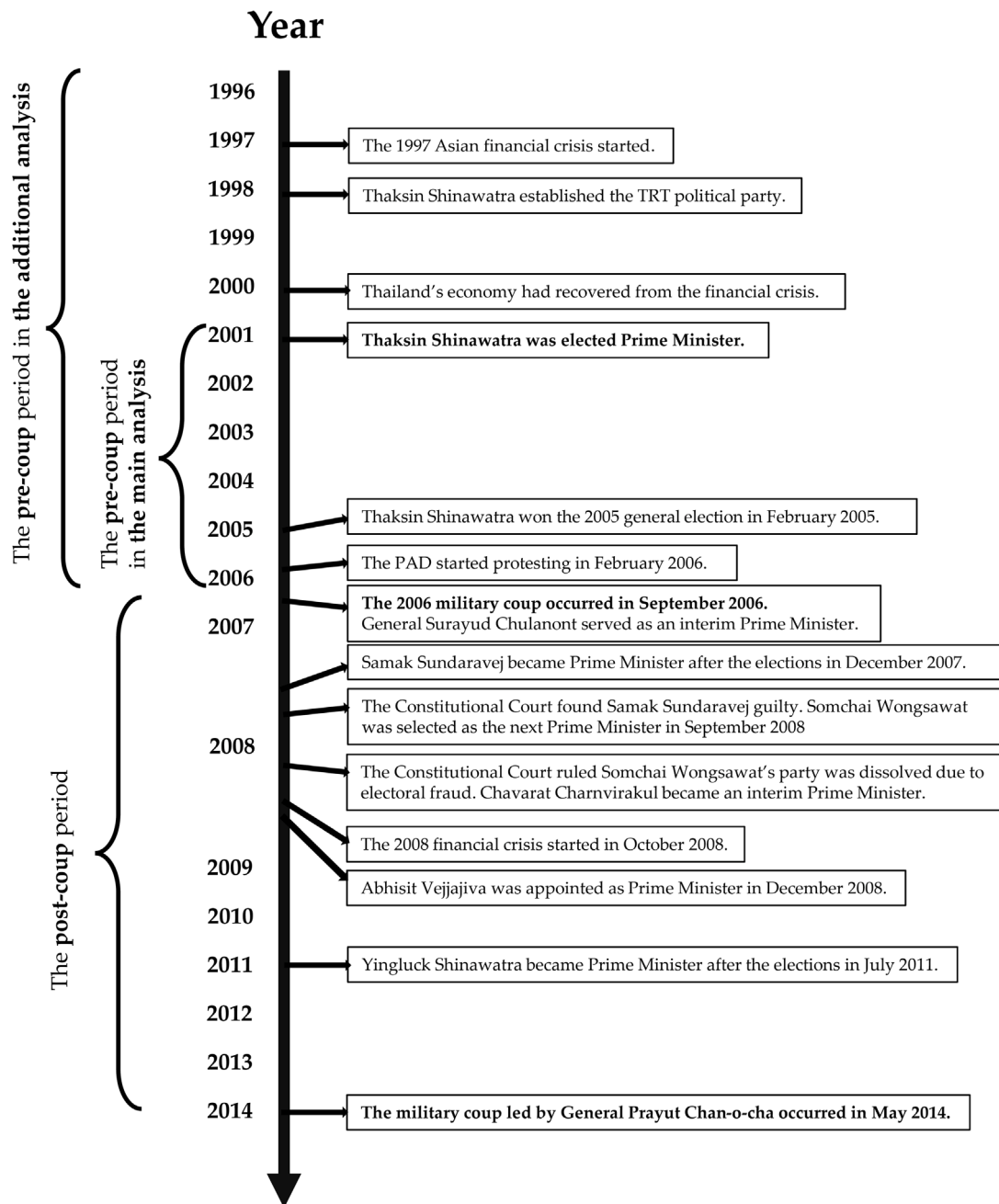


Fig. 1. A timeline of Thailand's politics from 1997 to 2014.

Surayud Chulanont's government planned a budget deficit of 147 billion Baht in 2007 and 168 billion Baht in 2008 while increasing the military expenditure by 35% in 2007. There was an effort to employ capital controls to deflate the Thai Baht following its significant appreciation, but this produced fear in the financial markets and a crash in the Thai stock market within a day. Other policies include fully freeing the 30-baht universal healthcare program and debt-relief measures for farmers.

Surayud Chulanont's economic policy was primarily centered on exports, and Thaksin Shinawatra's "dual-track policy", which intended to stimulate domestic demand while also attracting foreign investment, was abandoned. Many additional economic policies established by Thaksin Shinawatra's government were overturned, for example, subsidies for rice farmers, the Million Cows project, the One Laptop Per Child (OLPC) program, and personal computers and internet connections in all public and secondary schools.

Surayud Chulanont scheduled a referendum on the constitution for August 19, 2007, and general elections for December 2007. Samak Sundaravej led the People's Power Party (PPP) to serve as a proxy for Thaksin Shinawatra and pledged to continue Thaksin

Shinawatra's populist programs. On January 28, 2008, Samak Sundaravej became the prime minister of Thailand. After Samak Sundaravej launched amnesty for Thaksin Shinawatra and for banned TRT politicians, the PAD ("Yellow Shirts") organized protests. The Yellow Shirts and the Red Shirts confronted one another on numerous occasions, including in Udon Thani, Chiang Mai, and Bangkok. On August 26, the PAD took over the grounds of Government House. Tensions reached such a level that Samak Sundaravej announced a state of emergency in Bangkok on September 2.

Meanwhile, Samak Sundaravej was accused of violating the law forbidding government ministers from holding other paid employment when he was on two cooking shows while in office. The Constitutional Court found Samak Sundaravej guilty and obliged him to resign as prime minister. On September 9, 2008, Somchai Wongsawat, Thaksin Shinawatra's brother-in-law, was selected as the next prime minister by the majority of his party and coalition partners.

On December 2, 2008, the Constitutional Court ordered the People's Power Party and other coalition parties to be dissolved for electoral fraud. As a result, Wongsawat was also removed from office and was banned from politics. Chavarat Charnvirakul served as Thailand's acting prime minister for a brief time. Later, Abhisit Vejjajiva was appointed prime minister following a vote in parliament on December 15, 2008. The UDD ("Red Shirts") protested again in April 2009 and March–May 2010. The incidents included a declaration of a state of emergency, the deaths of protestors, and the destruction of a large shopping mall in Bangkok.

After the general elections in Thailand on July 3, 2011, Yingluck Shinawatra, Thaksin Shinawatra's youngest sister, became the first female prime minister in the history of Thailand. In 2013, political tensions were exacerbated by mass protests against her administration.

On May 22, 2014, General Prayut Chan-o-cha launched another successful military coup and has served as Thailand's prime minister since then.

The 2006 military coup thus marked the beginning of the collapse of Thailand's democratic regime.

3. The expected effects of the 2006 military coup on Thailand's economy

The effects of the 2006 military coup on Thailand's economy can be divided into (i) the direct short-term effects from the coup and (ii) the long-term effects of the coup's policies.

First, the direct short-term effects were arguably ambiguous. On the one hand, military coups in other countries' histories were associated with switching from a democratic regime to a military dictatorship regime by force. Therefore, one would expect the Thai economy to have experienced a large decrease in GDP, lower economic growth, lower investment, and perhaps sanctions from other countries. On the other hand, the 2006 military coup was somewhat anticipated, as the Yellow Shirts pressured the military to stage a coup. The business sector did not see the coup as the beginning of political instability but rather as a means of putting an end to political tensions and uncertainties. Therefore, one could argue that the coup would have returned the Thai economy to normal.

Second, the long-term effects of the coup's policies depend on whether Thaksin Shinawatra's populist policies were effective and, if they were effective, whether his policies were continued. Thaksin Shinawatra adopted similar economic policies to Venezuelan President Hugo Chavez, giving him the nickname the "Asian Chavez" (Phongpaichit and Baker, 2008). Social programs and economic policies, such as the debt-restructuring plan, the one-million-baht community fund, the subsidized universal health care program, the "One District, One Scholarship" project, and several government-funded infrastructure projects, significantly improved people's lives and increased the number of Thaksin Shinawatra's supporters (Pye and Schaffar, 2008).

However, Thaksin Shinawatra's economic plans had drawbacks and were not as effective as anticipated. For example, Pachanee and Wibulpolprasert (2006) discuss the drawbacks of the 30-baht universal health program that could offset its benefits. When the 30-baht universal health program was introduced, Thaksin Shinawatra also launched the medical hub project, in which foreign patients travel to Thailand for medical treatment. At that time, public hospitals were underfunded, and there was an "internal brain drain", in which medical doctors from public hospitals moved to private hospitals. Furthermore, Thaksin Shinawatra's economic plans benefited his network group. Capital would have been increasingly concentrated in the new elite group, and the wealthy grew wealthier.

Theoretically, although Thaksin Shinawatra's policies were effective and improved Thailand's economy significantly, the long-term effects of the coup could disappear if Thaksin Shinawatra's policies were continued. That is, the negative effects of the 2006 coup would appear only if Thaksin Shinawatra's policies were abandoned.

In reality, the military coup left the economy untouched, since it had no intention of seizing power from the people and only replaced the leader. The military leaders also appointed technocrats to keep the economy operating smoothly. Although the military government, led by General Surayud Chulanont, reversed several of Thaksin Shinawatra's economic policies, the government also introduced similar policies. Furthermore, between 2006 and 2014, the other prime ministers were too focused on politics to implement economic plans or reverse Thaksin Shinawatra's policies. For example, Samak Sundaravej's government appeared to stimulate the economy through growth-friendly policies but lacked clear direction or economic policy leadership (Pongsudhirak, 2008). Nonetheless, military spending was projected to increase following the coup, as General Surayud Chulanont had approved the increase.

Several countries, including Canada, the United States, Japan, South Korea, Malaysia, the United Kingdom, and European countries, reacted negatively to the military coup; they issued official statements condemning Thailand's military government and advising their citizens not to travel to Thailand, for safety reasons. Therefore, this study investigates the consequences of the military coup for international tourism and financial aid from the United States.

The coup's impact on international tourism was expected to be negative, particularly in the short term, due to the temporary travel warnings issued by other countries, which advised their citizens against non-essential travel to Thailand, citing concerns over safety and security. However, it is important to note that these travel warnings were only temporary and were lifted as soon as a fresh election was scheduled and the situation in Thailand stabilized. As a result, it is reasonable to expect that international tourism in Thailand would eventually return to its pre-coup level in the long term, once the political situation was resolved and the country became free of political tensions. Tourists were able to visit Bangkok and other destinations in Thailand without any significant risks to their safety or security.

The impact of the coup on financial aid from the United States was complex. The United States had a healthy relationship with the Thai military. The United States did not want to lose Thailand to communist advances during the Cold War (Farrelly, 2013). After the Vietnam War, the US military prioritized Thailand as a support for regional stability (Chambers, 2013). Furthermore, the US military regularly holds Cobra Gold military exercises in Thailand. After the military coups in 2006 and 2014, the United States reduced its involvement in Cobra Gold exercises, but they restarted in 2017 in an attempt to re-establish the relationship between Thailand and the United States.

In 2006, the relationship between Thailand and the United States became complicated. The United States' influence in Southeast Asia was endangered after Thailand's military chief visited China and the Chinese government assisted Thailand with much larger funds. The United States was forced to make a choice between imposing sanctions on the military government of Thailand and maintaining its influence in Southeast Asian affairs. Therefore, the impact of the coup on financial aid from the United States could be either negative (due to sanctions) or neutral (due to the attempt to maintain its relationship with Southeast Asia).

4. The synthetic control method

This paper aims to predict how Thailand would have been if the coup in 2006 had not occurred. The analysis investigates the impacts on six groups of economic variables: (i) GDP and inflation, (ii) components of GDP, (iii) the labor market, (iv) financial stability, (v) military expenditure, and (vi) international tourism and bilateral aid flows from the United States.

Since the author cannot go back in time to change real events, predicting the outcome of a counterfactual situation is challenging. Ideally, a counterfactual should match the behavior of actual Thailand in the years prior to 2006 but without experiencing a coup in 2006.

4.1. Methodology

To estimate the outcome of interest, in the absence of the treatment, I use the synthetic control method, a novel and transparent methodology for case studies, developed by Abadie and Gardeazabal (2003) and extended by Abadie et al. (2010). This method compares the actual outcome in the treated country with the weighted average of the outcome for all units in an untreated control group, i.e., countries wherein no coup happened in the period of interest.

Suppose that there are J countries in the control group. Each country is indexed by $j \in \{1, 2, \dots, J\}$. Thailand is indexed by 0. Let $Y_{0,t}$ denote the variable of interest of Thailand in year t and $Y_{j,t}$ denote the variable of interest of country j in year t . Let k denote the number of years before the coup, Y_0 be a $k \times 1$ vector of the pre-coup values of a variable of interest for Thailand, and Y_{cg} be a $k \times J$ matrix denoting the values of the same variables for the countries in the control group. A synthetic control is defined as a $J \times 1$ vector $W = [w_1, \dots, w_J]'$ that describes weights for the countries in the control group, such that $w_j \geq 0$ for all $j \in \{1, 2, \dots, J\}$ and $w_1 + w_2 + \dots + w_J = 1$.

The difference between the pre-coup values of Thailand and a synthetic control unit in the pre-coup period is described by the vector $Y_0 - Y_{cg}W$. The synthetic control is constructed by choosing the weight vector W^* that minimizes the norm of the difference vector¹:

$$\min_W (Y_0 - Y_{cg}W)'(Y_0 - Y_{cg}W).$$

After the optimal weight vector W^* is obtained, synthetic Thailand is constructed as the linear combination of the values of the countries in the control group with the optimal weight vector W^* :

$$\widehat{Y}_{0,t} = Y_{cg}W^* = \sum_{j=1}^J w_j^* Y_{j,t}.$$

The goodness of fit can be measured by the mean squared prediction error (MSPE) as

$$MSPE = \sum_{t=1}^k \left(Y_{0,t} - \sum_{j=1}^J w_j^* Y_{j,t} \right)^2.$$

The treatment effect estimated by the synthetic control method is represented by the difference between the values of actual Thailand $Y_{0,t}$ and synthetic Thailand $\widehat{Y}_{0,t}$.

$$Y_{0,t} - \widehat{Y}_{0,t} = Y_{0,t} - \sum_{j=1}^J w_j^* Y_{j,t}.$$

¹ The estimation is done using command `synth` on Stata (Abadie et al., 2011).

One underlying assumption is that the coup in Thailand has no effect on the countries in the control groups, and any changes in the outcome variable after period k are interpreted as having been caused by the coup.²

After the estimated effects are obtained, a natural question is how to test for statistical significance. Abadie et al. (2010) point out that the synthetic control method uses a small number of units, but statistical techniques often consider asymptotic properties. Therefore, a common practice is to conduct falsification exercises called *placebo tests*. The intuition is that a placebo test conducts the same synthetic control method for all countries in the control group and compares the magnitudes of the effects in their synthetic controls with the magnitude of the effect in Thailand. The estimated effect in Thailand should be large relative to the distribution of the estimated effects in the countries in the country group that are not exposed to the coup.

Following the idea of the placebo test in Abadie et al. (2010), each experiment is accompanied by placebo tests showing the predicted effects of countries in the country group that have a pre-coup MSPE of smaller than five times the pre-coup MSPE of Thailand. In addition, because the 95% confidence intervals cannot be calculated directly, I follow Abadie et al. (2010) by providing the information on the 5th percentile and the 95th percentile of the predicted effects from the placebo tests. The estimated gap of Thailand can be compared with the distribution of estimated gaps of other countries whose effects should be zero.³ When a gap of Thailand is below the 5th percentile, the probability that the gap of Thailand is negative under the null hypothesis that the true effect is zero is less than 5%. Similarly, when a gap of Thailand is above the 95th percentile, the probability that the gap of Thailand is positive under the null hypothesis that the true effect is zero is less than 5%.

For some variables, there are some placebo tests of countries that cannot be analyzed because of technical errors in the calculation; for example, their Hessian matrices are unstable or asymmetric, or Stata cannot calculate their numerical derivatives.

4.2. Data

The analysis concerns six groups of economic variables: (i) GDP and inflation, (ii) components of GDP, (iii) labor market, (iv) financial stability, (v) military expenditure, and (vi) international tourism and bilateral aid flows from the United States. The data were downloaded from World Bank Open Data.⁴

The first group includes real GDP, real GDP per capita, GDP growth rate, and CPI inflation rate. Real GDP and real GDP per capita are expressed in logarithms. The second group includes consumption, investment, government expenditure, exports, imports, and trade openness. All of these variables are described as a percentage of GDP. Trade openness is measured as the sum of exports and imports as a percentage of GDP. The third group has the labor force participation rate and the unemployment rate. The labor force participation rate is the percentage of the population ages 15 and above working or actively looking for a job. The unemployment rate is the percentage of the labor force that does not have a job but is actively searching for employment.

The fourth group includes government debt, total reserves, and total reserves minus gold. The fifth group has military expenditure, military expenditure as a percentage of GDP, and military expenditure as a percentage of government expenditure. The sixth group contains the number of international tourists, the income from international tourists, and the bilateral aid flows from the United States. The number of international tourists is the number of international tourists who travel to a country other than that in which they have established their residency. The income from international tourists is measured by the spending of international visitors, including expenditure on domestic carriers for international transport. Bilateral aid flows are defined as the net disbursements of official development assistance or official aid from the members of the Development Assistance Committee (DAC).

The process of selecting countries for the control group is a critical issue. Abadie and Gardeazabal (2003) argue that the countries chosen should behave similarly to the country with the intervention. Abadie et al. (2015) emphasize that including irrelevant countries could lead to interpolation biases and overfitting. To capture similarities in culture, economy, and regional co-movements, I use only Southeast Asian countries and Thailand's large trading partner countries.⁵ The control group consists of the following 26 countries: Australia, Bangladesh, Brunei, Cambodia, Canada, China, France, Germany, Hong Kong, India, Indonesia, Japan, South Korea, Lao People's Democratic Republic, Malaysia, Myanmar, Netherlands, New Zealand, the Philippines, Russia, Saudi Arabia, Singapore, Switzerland, the United Kingdom, the United States, and Vietnam. Table 1 provides the full list of countries, with the country codes, regions, and income groups. Column "Subgroup" indicates whether a country is included in the 26-country subgroup.

In the main analysis, I use data from 2001 to 2014. The data is separated into the pre-coup period between 2001 and 2005 and the post-coup period between 2006 and 2014. I select the pre-coup period of 2001–2005 to match the period in which Thaksin Shinawatra served as prime minister because it is likely that Thaksin Shinawatra's new economic policies could cause a structural change in Thailand's economy in 2001. In addition, Thailand experienced a financial crisis in 1997, the so-called Tom Yam Kung crisis, which caused a large decrease in GDP in countries in East Asia and Southeast Asia, especially South Korea, Indonesia, Hong Kong, Laos, Malaysia, and the Philippines. Thailand's GDP growth rate was -2.8% in 1997 and -7.6% in 1998 before it recovered in 1999 and 2000. Therefore, the pre-coup period starts from 2001, when Thaksin Shinawatra started his first term and Thailand's economy returned to its long-run level.

² Cao and Dowd (2019) suggest an adjustment for a synthetic control method with spillover effects.

³ This study's analysis can be considered as a set of separate hypothesis tests since each experiment estimates its unique weights for creating a synthetic Thailand and performs an independent placebo test. The specifications do not suffer from multiple hypothesis testing problems.

⁴ <https://data.worldbank.org/>

⁵ Thailand is a small open economy that is vulnerable to external shocks through its trade network. See Apaitan et al. (2022), Bhattarai et al. (2020), Yamamoto (2014), and Yang and Hamori (2014) for examples.

Table 1

The list of the countries in the analysis.

Country	Country code	Region	Income	Subgroup	Country	Country code	Region	Income	Subgroup
Albania	ALB	2	4		Kazakhstan	KAZ	2	4	
Algeria	DZA	4	4		Korea, Rep.	KOR	1	1	Y
Armenia	ARM	2	4		Lao PDR	LAO	1	3	Y
Australia	AUS	1	1	Y	Lebanon	LBN	4	4	
Azerbaijan	AZE	2	4		Libya	LBY	4	4	
Bangladesh	BGD	6	3	Y	Malaysia	MYS	1	4	Y
Belarus	BLR	2	4		Maldives	MDV	6	4	
Belize	BLZ	3	4		Marshall Islands	MHL	1	4	
Bosnia & Herzegovina	BIH	2	4		Mauritius	MUS	7	4	
Botswana	BWA	7	4		Mexico	MEX	3	4	
Brazil	BRA	3	4		Montenegro	MNE	2	4	
Brunei	BRN	1	1	Y	Myanmar	MMR	1	3	Y
Bulgaria	BGR	2	4		Namibia	NAM	7	4	
Cambodia	KHM	1	3	Y	Netherlands	NLD	2	1	Y
Canada	CAN	5	1	Y	New Zealand	NZL	1	1	Y
China	CHN	1	4	Y	North Macedonia	MKD	2	4	
Colombia	COL	3	4		Paraguay	PRY	3	4	
Costa Rica	CRI	3	4		Peru	PER	3	4	
Cuba	CUB	3	4		Philippines	PHL	1	3	Y
Dominica	DMA	3	4		Romania	ROU	2	4	
Dominican Rep.	DOM	3	4		Russia	RUS	2	4	Y
Ecuador	ECU	3	4		Samoa	WSM	1	4	
Guinea	GNQ	7	4		Saudi Arabia	SAU	4	1	Y
Fiji	FJI	1	4		Serbia	SRB	2	4	
France	FRA	2	1	Y	Singapore	SGP	1	1	Y
Gabon	GAB	7	4		South Africa	ZAF	7	4	
Germany	DEU	2	1	Y	St. Lucia	LCA	3	4	
Grenada	GRD	3	4		St. Vincent	VCT	3	4	
Guatemala	GTM	3	4		Suriname	SUR	3	4	
Guyana	GUY	3	4		Switzerland	CHE	2	1	Y
Hong Kong, China	HKG	1	1	Y	Thailand	THA	1	4	Y
India	IND	6	3	Y	Tonga	TON	1	4	
Indonesia	IDN	1	3	Y	Turkey	TUR	2	4	
Iran	IRN	4	4		Turkmenistan	TKM	2	4	
Iraq	IRQ	4	4		United Kingdom	GBR	2	1	Y
Jamaica	JAM	3	4		United States	USA	5	1	Y
Japan	JPN	1	1	Y	Venezuela, RB	VEN	3	4	
Jordan	JOR	4	4		Vietnam	VNM	1	3	Y

Region: 1 = East Asia & Pacific, 2 = Europe & Central Asia, 3 = Latin America & Caribbean, 4 = Middle East & North Africa, 5 = North America, 6 = South Asia, 7 = Sub-Saharan Africa.

Income: 1 = High income, 2 = Low income, 3 = Lower middle income 4 = Upper middle income.

Income: Y = included in the 26 countries in the main control group.

This paper considers the post-coup period of 2006–2014 because General Prayut Chan-o-cha led another coup on 22 May 2014. Moreover, because the period used in this paper overlaps with the 2008 global financial crisis, the interpretation for the years after 2008 should be discussed with caution.

For robustness, I consider two additional analyses in which the pre-coup period and the set of countries in the control groups are extended. In the first additional analysis, called R#1, I extend the pre-coup period to the period between 1996 to 2006. The idea is that synthetic Thailand constructed from a longer time pre-coup period may better capture the long-term trend of actual Thailand and is less sensitive to short-term noises. There are two disadvantages of using the extended pre-coup period. First, there could be complications from the 1997 financial crisis, which significantly affected Thailand and other countries in the control group. The estimation from the synthetic control method may inaccurately put more weights toward countries that were largely affected by the financial crisis. Second, due to the lack of data availability in the early years, some variables have a smaller number of countries in the control group.

To surpass the data availability problem, in the second additional analysis, called R#2, I also add 49 upper-middle-income countries to the control group. Table 1 provides the list of all 75 countries in the control group. These additional upper-middle-income countries shall be similar to Thailand in terms of the dynamic paths of economic activities and developments. One advantage of this analysis is that we can show that the empirical results are not sensitive to the choice of countries in the control group.⁶ One consideration is that a larger control group creates a substantial risk of overfitting and may cause additional noisy predictions (Abadie et al., 2015).

⁶ That is, I impose minimal parameter restrictions to prove that I did not manipulate the results for political purposes.

4.3. The 2008 global financial crisis

In this section, I discuss how Thailand experienced the global financial crisis in 2008, and then I discuss why the synthetic control method used in this paper is not significantly affected by the financial crisis.

The 2008 financial crisis significantly impacted the global economy, leading to widespread job loss, business failures, and a severe recession. The crisis began in the United States and quickly spread to other countries due to the interconnectedness of the global financial system. Many countries experienced significant declines in GDP, including the United States, which saw a decline of 2.8% in 2009. Unemployment rates also rose significantly, with many countries experiencing double-digit unemployment rates. The crisis also led to a decline in consumer spending and business investment, which further exacerbated the economic downturn.

As the global crisis started outside Thailand in 2008, its effects were channeled into the Thai economy through the demand for Thai exports. The effect of the 2008 financial crisis on Thailand's economy seems to be a temporary negative shock. GDP growth rate was -0.7 percent in 2009 before it recovered to 7.5 percent in 2010. CPI inflation was -0.8 percent in 2009 and increased to 3.2 percent in 2010. Consumption and investment as percentages of GDP declined, and government expenditure as a percentage of GDP increased. Exports and imports fell in 2009, making overall trade openness fell as well. Thailand's economic indicators experienced a temporary negative shock in 2009 and recovered to normal in 2010.

The 2008 financial crisis took the world by surprise, catching many experts and governments off guard. Despite a few warning signs, such as the subprime mortgage crisis in the United States, the scale and scope of the crisis were not fully understood until it was already underway. Governments and financial institutions around the world were left scrambling to respond, and the long-term effects of the crisis are still being felt in many countries today. This was truly an unprecedented event that was unlikely to have been predicted by any single country or institution.

When it comes to Thailand specifically, there were several factors that contributed to the country's lack of anticipation of the financial crisis. One of the main reasons is that Thailand's financial institutions were not as intertwined with the global financial system as those in some other countries. As a result, Thai policymakers did not have the same level of exposure to the risk factors that were present in other parts of the world. Furthermore, Thailand had experienced a major financial crisis in 1997, which resulted in significant structural changes to the country's financial system. This crisis led to the establishment of new regulations and institutions designed to prevent a similar situation from occurring in the future.

Another reason why Thailand was unlikely to have anticipated the 2008 financial crisis is related to the country's political and economic situation at the time. The 2006 military coup in Thailand resulted in a period of political instability and uncertainty. This made it difficult for policymakers to focus on anything other than stabilizing the country's political situation. Additionally, Thailand's economy was performing well during this time, with strong GDP growth and low inflation.

One natural question is whether countries in the control group were affected by the financial crisis in the same way Thailand's economy was. Fig. 2 shows a comparison of the GDP growth rates and inflation rates of Thailand and the countries in the control groups. We can see that most countries experienced a significant decrease in 2009 and rebounded in 2010. In addition, because countries in the control group should be able to represent Thailand, the main analysis uses only the countries in Southeast Asia and Thailand's primary trading partners. The Southeast Asian countries share similarities with Thailand regarding the level of development and the economic environment. Thailand's primary trading partner can capture the market effect that drives Thailand's exports and imports, as well as other economic variables.

5. The effects of Thailand's 2006 coup

This section presents results from the synthetic control method. I construct a synthetic control for each of the variables. Each analysis may use a different subset of countries because no information on the outcome variable should be missing for countries in the control group. The summary statistics of the variables of interest among the countries in the control group and the weights of the countries in the control group used to construct synthetic Thailand are provided in the online appendix.

I show the effects on six groups of economic variables: (i) GDP and inflation, (ii) components of GDP, (iii) labor market, (iv) financial stability, (v) military expenditure, and (vi) international tourism and the bilateral aid flows from the United States. The empirical results from the main analysis are shown in Figs. 3–9.

For each variable, I provide two plots. The first graph compares the values of the variable between 1996 and 2014 for actual Thailand and three synthetic Thailands. The solid black line represents actual Thailand, the solid gray line represents synthetic Thailand based on the main analysis, the dashed gray line represents synthetic Thailand based on the additional analysis R#1, and the dotted gray line represents synthetic Thailand based on the additional analysis R#2.

The values of synthetic Thailand for the years 1996–2005 show how well the synthetic control method can replicate actual Thailand in the period before the 2006 coup, and the values of synthetic Thailand for the years 2006–2014 are counterfactual values that represent how Thailand would have been if the coup had not happened.

The second plot presents the estimated effects and their corresponding confidence intervals calculated from the placebo tests. The gaps and the confidence intervals from the main analysis are represented by the solid black line and the dashed black line. The gaps and the confidence intervals from the additional analysis R#1 are represented by the solid dark gray line and the dashed dark gray line, while those from the additional analysis R#2 are represented by the solid light gray line with dots and the dashed light gray line with dots.

The online appendix is divided into three sections; each section corresponds to a different analysis and provides estimation results in detail. There are three graphs for each variable. The first graph compares actual Thailand and synthetic Thailand. The second graph presents the placebo tests, which compare the estimated gaps from all countries in the control group. The third graph presents the gap between actual Thailand and synthetic Thailand and the corresponding confidence interval.

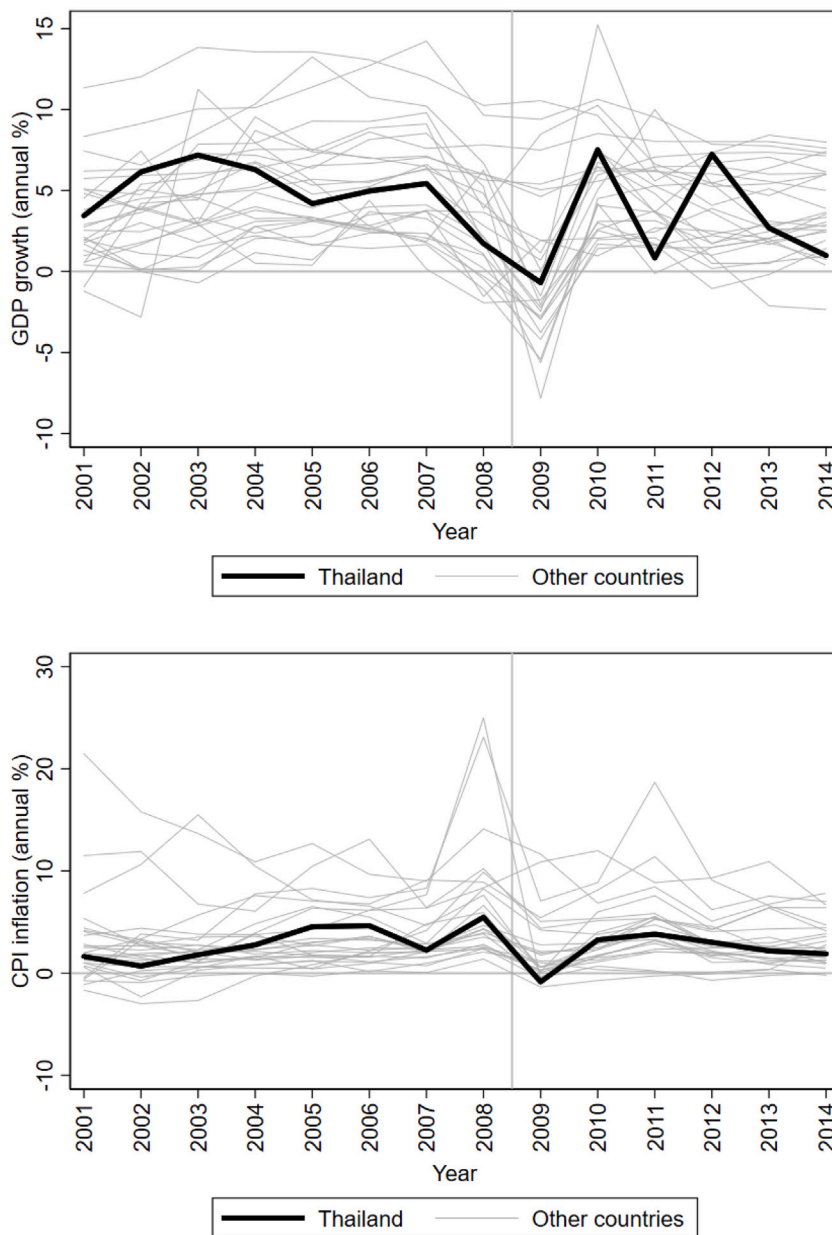


Fig. 2. GDP growth rates and inflation rates around the 2008 financial crisis.

5.1. The effects on the economy

5.1.1. Overview of the economy: GDP and inflation

Fig. 3 compares real GDP, real GDP per capita, the GDP growth rate, and inflation in actual Thailand and synthetic Thailands.

Real GDP and the real GDP per capita of actual Thailand and synthetic Thailand co-move closely in 2001–2005, but real GDP and the real GDP per capita of actual Thailand are consistently smaller than those of synthetic Thailands every year after 2006. Synthetic Thailands from the main analysis and the R#1 analysis are similar. In the main analysis, the gap of real GDP is equal to -2.1 , -1.7 , and -3.5 percent in 2006, 2007, and 2008, respectively, and fluctuates between -4.6 and -8.3 percent in 2009–2014. The case of real GDP per capita is similar to the case of real GDP; the gap is -2.7 percent in 2006, -2.1 percent in 2007, and -2.4 percent in 2008 and then fluctuates between -2.1 and -6.8 percent from 2009–2014.

The GDP growth rate and the inflation rate of actual Thailand move closely to those of synthetic Thailands, and they are mostly smaller than those of synthetic Thailands in the years 2006–2009. Although gaps of GDP, GDP growth rates, and inflation rates

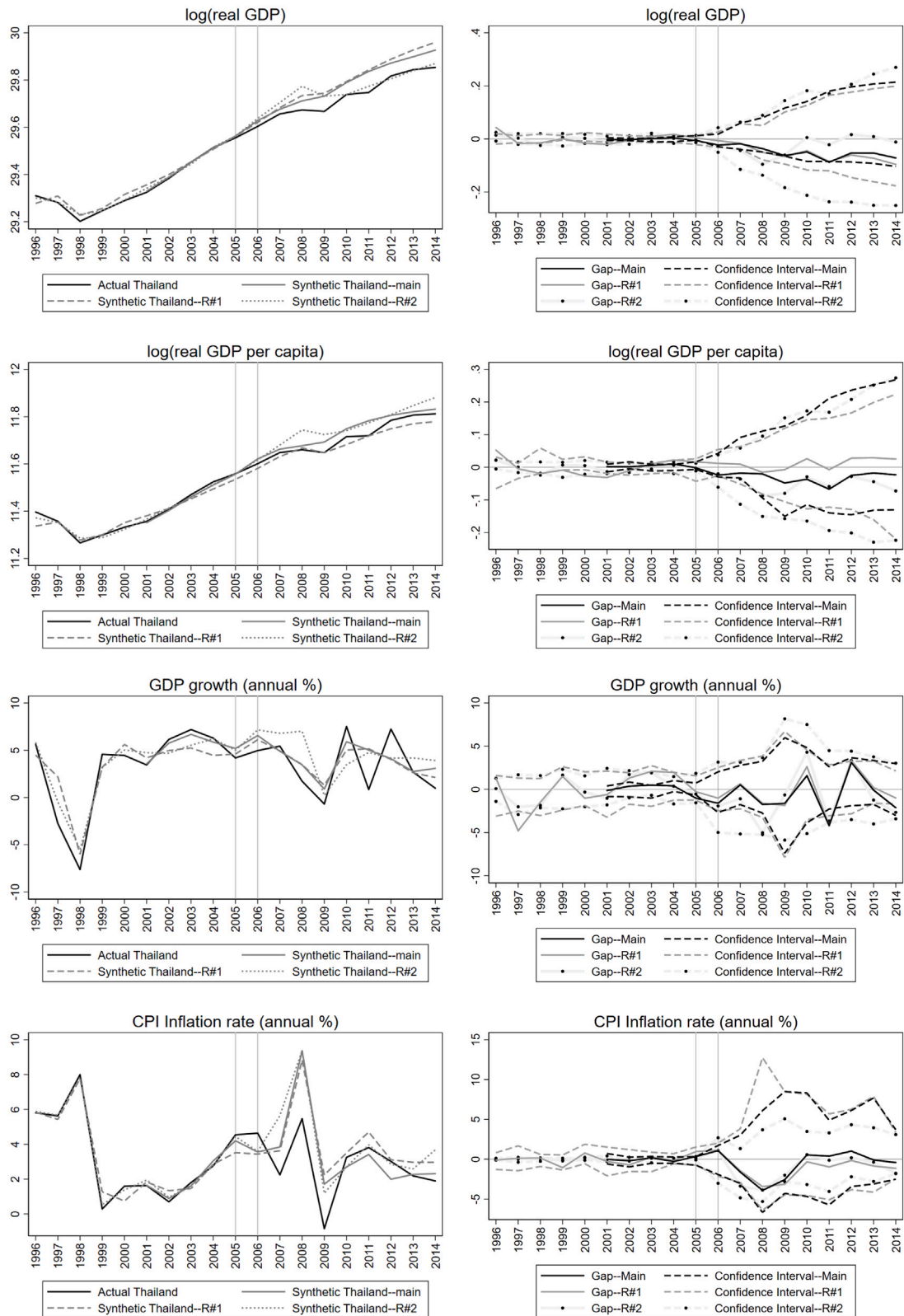


Fig. 3. Real GDP and real GDP per capita.

Note: The main analysis has 26 countries from 2001 to 2014. The additional analysis R#1 has 26 countries from 1996 to 2014. The additional analysis R#2 has 75 countries from 1996 to 2014.

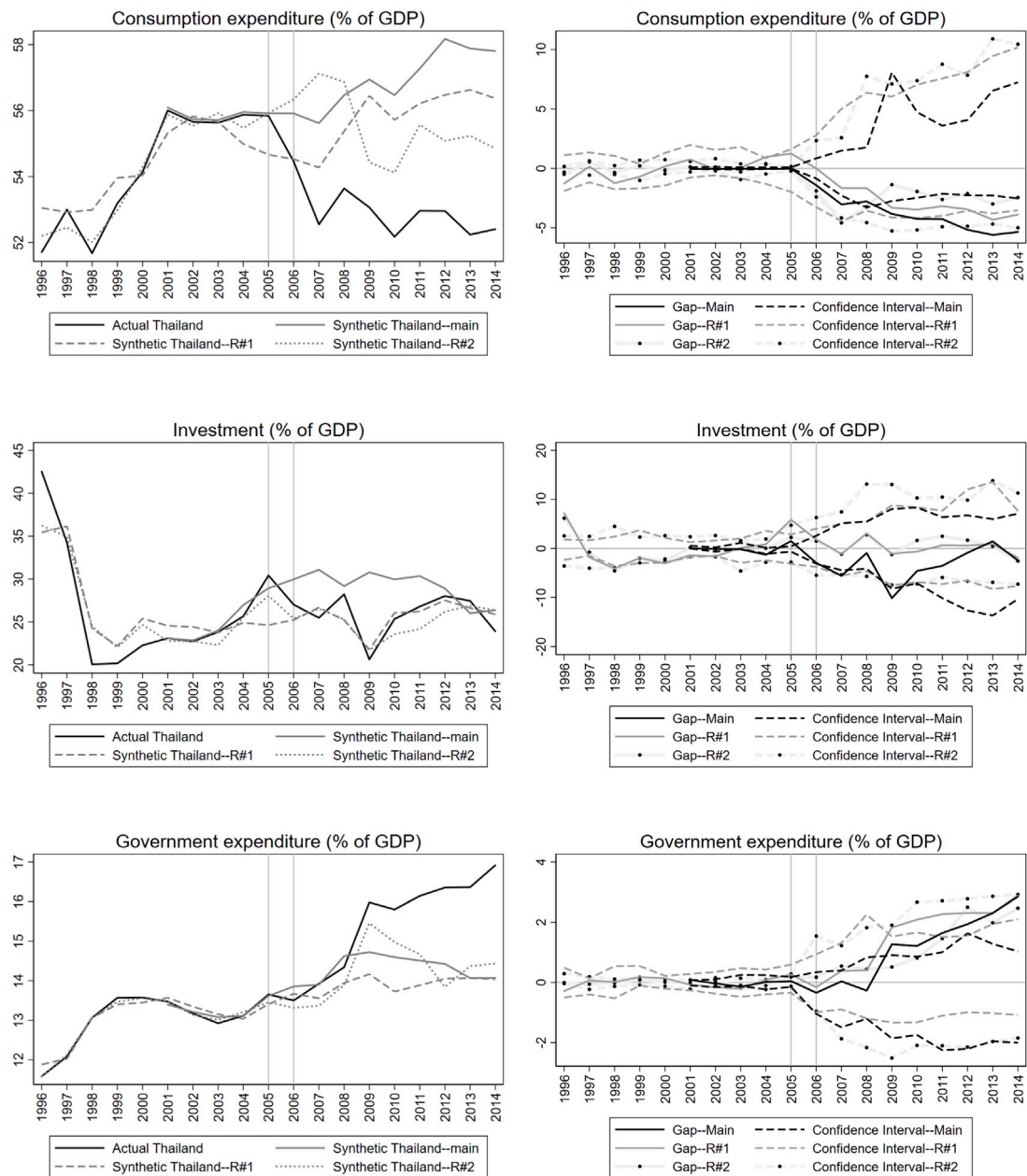


Fig. 4. Components of GDP part I: consumption, investment, and government expenditure.

Note: The main analysis has 26 countries from 2001 to 2014. The additional analysis R#1 has 26 countries from 1996 to 2014. The additional analysis R#2 has 75 countries from 1996 to 2014.

seem large, the placebo tests cannot conclude that the predicted gaps are statistically significant because the gaps lie between the 5th percentile line and the 95 percentile line.

The insignificance of the effect of the military coup on GDP is not surprising. As discussed in Section 3, several factors play a role at the same time. It could be that (i) Thaksin Shinawatra's populist policies were not effective, or (ii) Thaksin Shinawatra's policies were effective and subsequent prime ministers continued similar policies that also drove the economy. The synthetic control method does not detect that the economy before 2006 and after 2006 are structurally different.

5.1.2. Components of GDP: Consumption, investment, government spending, and net exports

Figs. 4 and 5 show the exercises on components of GDP: consumption, investment, government spending, and net exports.

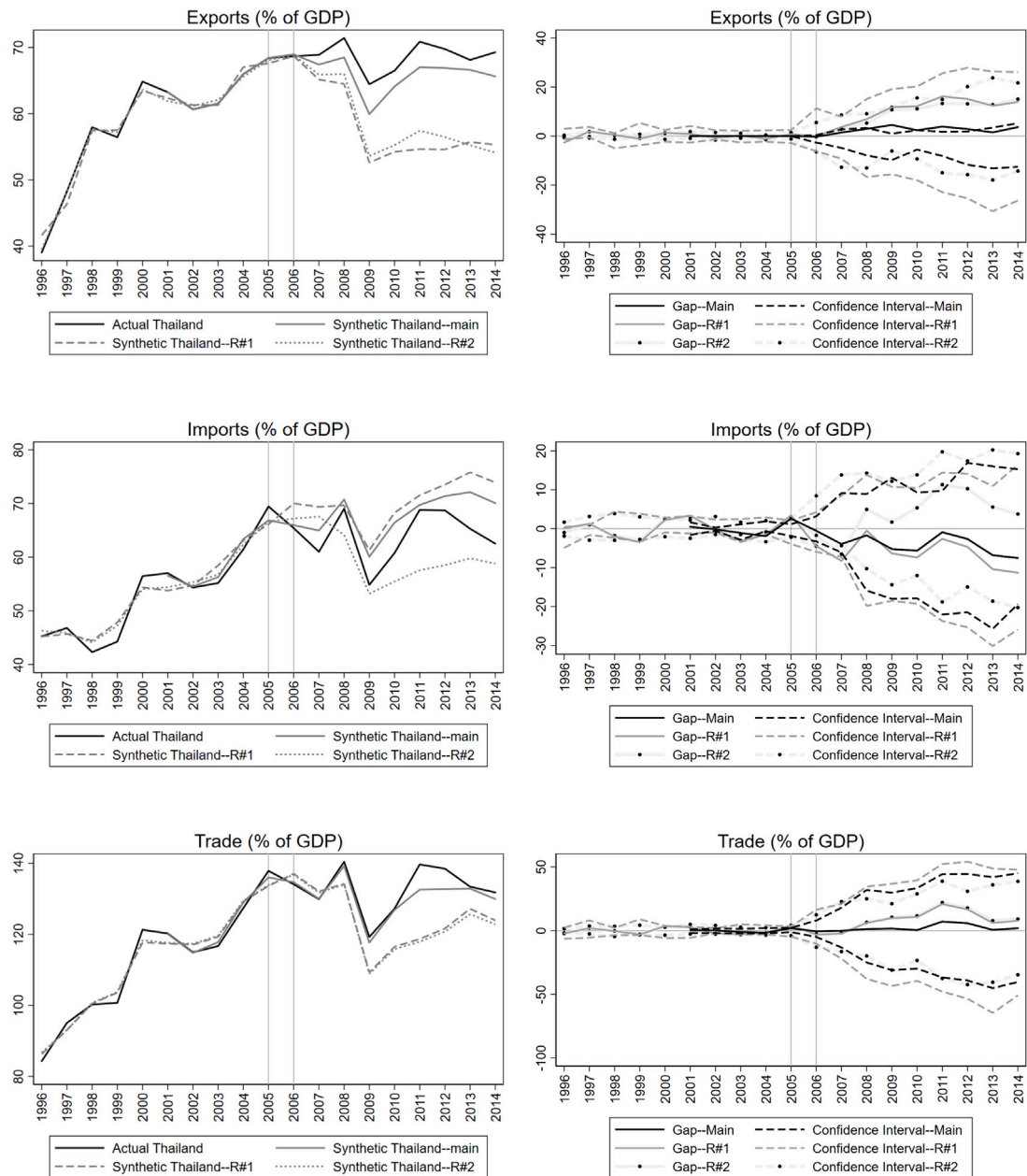


Fig. 5. Components of GDP part II: exports, imports, and trade openness.

Note: The main analysis has 26 countries from 2001 to 2014. The additional analysis R#1 has 26 countries from 1996 to 2014. The additional analysis R#2 has 75 countries from 1996 to 2014.

Consumption as a percentage of GDP decreases substantially after 2006; it is around 55.6–56.0 percent of GDP prior to 2006 and decreases to 52.2–53.6 percent of GDP in 2007–2014. Between 2005 and 2007, actual Thailand's consumption decreases by 3.3 percentage points. Synthetic Thailand from the main analysis can replicate actual Thailand in the pre-coup period well. Synthetic Thailand's consumption is relatively constant in 2001–2007, as it fluctuates in the range of 55.5–56.0 percentage points and increases to 56.9 percentage points in 2009 and 58.1 percentage points in 2012. The gap between actual Thailand and synthetic Thailand is –1.4, –3.0, –2.8, –3.8, –4.2, and –4.22 percentage points of GDP, respectively, in the years 2006–2011, before it jumps the range of –5.1 and –5.6 percentage points of GDP in 2012–2014. The placebo tests suggest that Thailand's gap of consumption as a percentage of GDP is significantly negative, compared to the gaps of other countries. This conclusion is supported by the additional analysis R#2.

Thanks to Thaksin Shinawatra's stimulus policies during 2001–2005, actual Thailand's investment has become a growing part of GDP. It increases from 23.1 percentage points in 2001 to 30.4 percentage points in 2005, then decreases to 25.5 percentage points in 2007, and falls further to 20.6 percentage points in 2009, after the financial crisis. Synthetic Thailand in the main analysis predicts that investment would have been consistently larger between 2006–2012 had the coup not occurred. However, synthetic Thailand from the additional analyses R#1 and R#2 predict that investment would have been similar to actual investment. Although the main analysis finds that the effect on investment is statistically significant in 2006, 2007, and 2009, the effects in the additional analyses are not statistically significant. Therefore, it is difficult to conclude a negative effect of the 2006 coup on Thailand's investment as a percentage of GDP.

Government expenditures of actual Thailand and synthetic Thailand move together in 2001–2008. After the financial crisis, government expenditure of actual Thailand increases from 14.3 percentage points of GDP in 2008 to 16.0 percentage points of GDP in 2009. Government expenditure of Synthetic Thailand from the main analysis slightly increases to 14.7 percentage points of GDP in 2009 and slowly declines to 14.1 percentage points of GDP in 2014. The gap is small between 2006–2008 and increases to 1.3 percentage points in 2009 and 2.9 percentage points of GDP in 2014. Synthetic Thailand from the analysis R#1 suggests larger gaps between 2009 and 2014, as the counterfactual government expenditure is relatively stable at around 14 percentage points of GDP.

The placebo tests suggest that Thailand's gap of government expenditure as a percentage of GDP is significantly positive from 2009–2014, according to the main analysis and the additional analysis R#1. Since the estimated effects occur after 2008, it could be that the synthetic control method captures the effect of the financial crisis. There are also other explanations. First, it could be that the coup allowed the government to immediately increase the budget as government stimulus policy in response to the crisis. Second, it could be that the decrease in consumption and investment as percentages of GDP increased the fraction of government expenditure in the GDP.

After 2006, exports (as a percentage of GDP) of actual Thailand are consistently larger than those of synthetic Thailand. Based on the main analysis, the gap of exports as a percentage of GDP is -0.27 percentage points in 2006 and increases to the range of 1.5–4.5 percentage points between 2008–2014, and the gap of imports as a percentage of GDP is -0.5 percentage points in 2006 and -3.9 percentage points in 2007 before it increases to -5.2 and -5.6 percentage points in 2009 and 2010. Having said that, the placebo tests cannot conclude that the effects on exports and imports are statistically significant. Actual Thailand and synthetic Thailand are open to international trade to approximately equal degrees. This is not surprising that the gap for trade openness is close to zero, because the gaps for exports and imports move in opposite directions.

In this section, we can conclude that the gap of consumption is significantly negative in the short run, and the gap of government expenditure is significantly positive after 2009.

5.1.3. Labor market

Traditionally, Thailand has had low unemployment rates, at around 0.5 to 2 percent. A common explanation is that the informal sector and the agricultural sector can quickly absorb unemployed workers.

According to Fig. 6, synthetic Thailand can match the labor force participation rate well. The labor force participation rates of synthetic Thailand stay in the range of 70–74 in 2006–2014, and the labor force participation rate of actual Thailand fluctuates between 72.8 and 73.1 before it drops to 70.6 in 2014. The placebo tests cannot detect a statistically significant change in the labor force participation rate.

Between the years 2001 and 2014, actual Thailand's unemployment rate appears to have gradually decreased overall, with a slight increase in 2009; the unemployment rate in 2001 was 2.6 percent, and by 2014 it had decreased to 0.6 percent. Synthetic Thailand from the analyses R#1 and R#2 are identical. Based on the main analysis, synthetic Thailand's unemployment rate decreases from 2.1 percent in 2001 to 1.3 percent in 2018 and then fluctuates in the range of 1.1–1.3 percent until 2014. The synthetic control method predicts that the gap is -0.3 percentage points in 2006 and -0.2 percentage points in 2007 before the gap is magnified to around -0.6 between 2010 to 2014.

The placebo tests cannot conclude that the gaps are statistically significant from zero.

5.1.4. Financial stability

Fig. 7 presents the coup's effect on Thailand's financial stability. Synthetic Thailand from the additional analyses R#1 and R#2 struggle to replicate the inverted-U shape of actual Thailand's data from 1996 to 2005. From 2005 to 2009, the values of government debt as a percentage of the GDP of actual Thailand and synthetic Thailand from the main analysis and the analysis R#2 move closely together. Synthetic Thailand from the main analysis has a large increase in government debt as a percentage of GDP after 2010. The gap widens from around -0.5 in 2007 and 2008 to -7.6 , -8 , and -16 between 2010 and 2012. Despite large effects, the placebo test cannot conclude that the effect on government debt is significant.

Actual Thailand has a large increase in the total reserves and the total reserves minus gold. Synthetic Thailand shows an upward trend as well, but the magnitude of the increase is not comparable to the graph of actual Thailand. The gap of the total reserves of actual Thailand and synthetic Thailand increases from 4.9 percent in 2006 to 41.5 percent in 2009 and then decreases back to 2.24 percent in 2012. Similarly, the gap of the total reserves minus gold of actual Thailand and synthetic Thailand increases from 15.4 percent in 2006 to 90.4 percent in 2010 and then decreases back to 29.5 percent in 2014. The placebo tests cannot detect statistically significant effects on the total reserves and the total reserves minus gold.

Previous works in the literature find that the 2006 coup affected the financial markets around the time of the coup (e.g., Khanthavit, 2019; Kongprajya, 2010.) This paper finds that the coup did not have a substantial impact on confidence in Thailand's financial markets on a worldwide scale. Both government debts and total reserves were similar to what they would have been if the coup had not occurred.

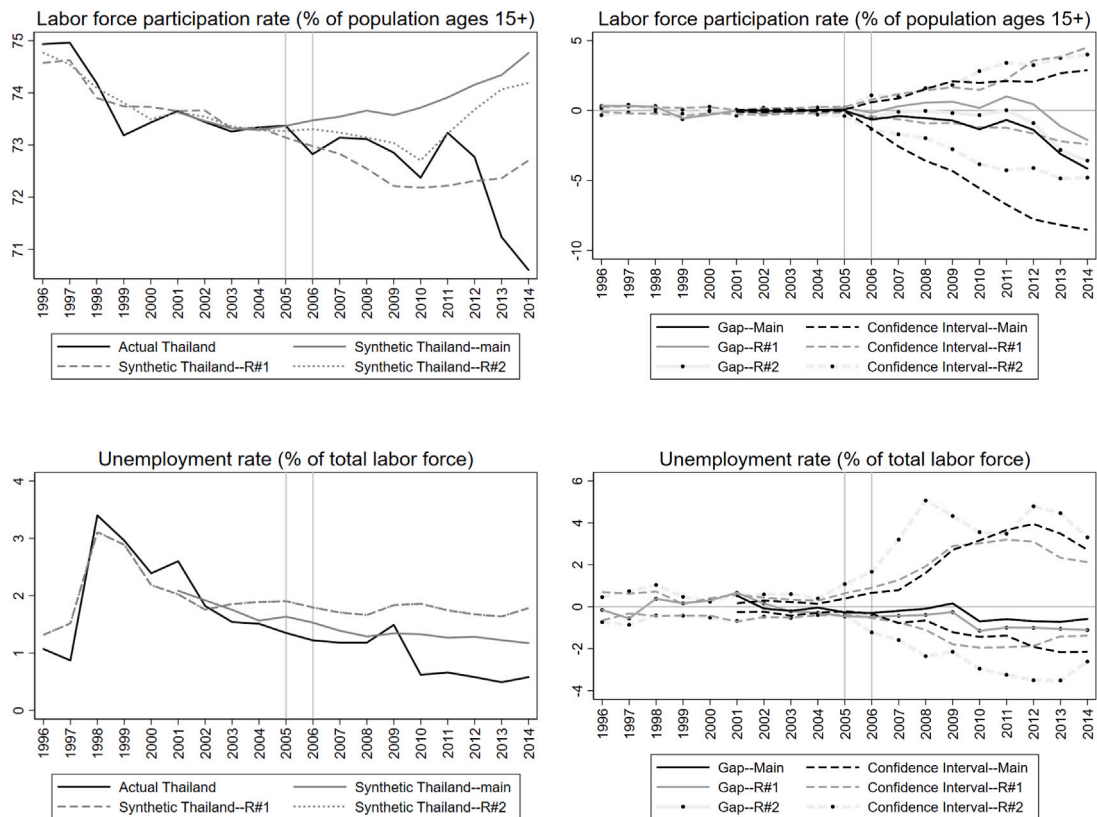


Fig. 6. The labor force participation rate and the unemployment rate.

Note: The main analysis has 26 countries from 2001 to 2014. The additional analysis R#1 has 26 countries from 1996 to 2014. The additional analysis R#2 has 75 countries from 1996 to 2014.

5.2. Results on military expenditure

This section presents the main results of the paper. I examine the impact of coups on military expenditure levels, military expenditure as a percentage of GDP, and military expenditure as a percentage of overall government expenditure. The analysis results are depicted in Fig. 8. Synthetic Thailand from the additional analysis R#1 and R#2 are identical, so they overlap on the figure. However, their confidence intervals are different.

Synthetic Thailand successfully matches actual Thailand in terms of the level of military expenditure and military expenditure as a percentage of GDP in 2001–2005. After 2006, synthetic Thailand has a steady increasing military expenditure and a decreasing military expenditure as a percentage of GDP. These trends continue from the pre-2006 period. In contrast to synthetic Thailand, actual Thailand has a large increase in military expenditure measured as the level and the percentage of GDP. Its military expenditure increases from 79 billion baht in 2005 (around 2.3 billion US dollars) to 165 billion baht in 2009 (around 4.8 billion US dollars).⁷ Equivalently, actual Thailand's military expenditure increases from 1.12 percent of GDP in 2005 to 1.82 percent of GDP in 2009.

Compared to the military expenditure of synthetic Thailand, the military expenditure of actual Thailand is larger by 13.1 percent in 2006, 16.1 percent in 2007, 72.5 percent in 2008, and around 78.3–102.6 percent from 2009 to 2014. In terms of the percentage of GDP, the gap is around 0.4–0.7 percentage points of GDP from 2007 to 2011 and around 0.3 from 2012 to 2014.

According to the placebo tests, the effects on military expenditure between 2007 and 2009 are statistically different from zero, and the effects on military expenditure as a percentage of GDP between 2007 and 2011 are statistically different from zero. This result is expected because, as explained in Section 3, General Surayud Chulanont, the prime minister after the military coup, approved a large increase in military expenditure. The increase is also consistent with the findings in the literature that military expenditure often increases after the country has a military coup.

In Fig. 8, I also consider military expenditure as a percentage of government expenditure to measure how the government allocated its budget. However, all synthetic Thailand fail to match actual Thailand. Therefore, this experiment is not discussed in this paper.

⁷ The exchange rate was around 34 Thai baht for 1 US dollar.

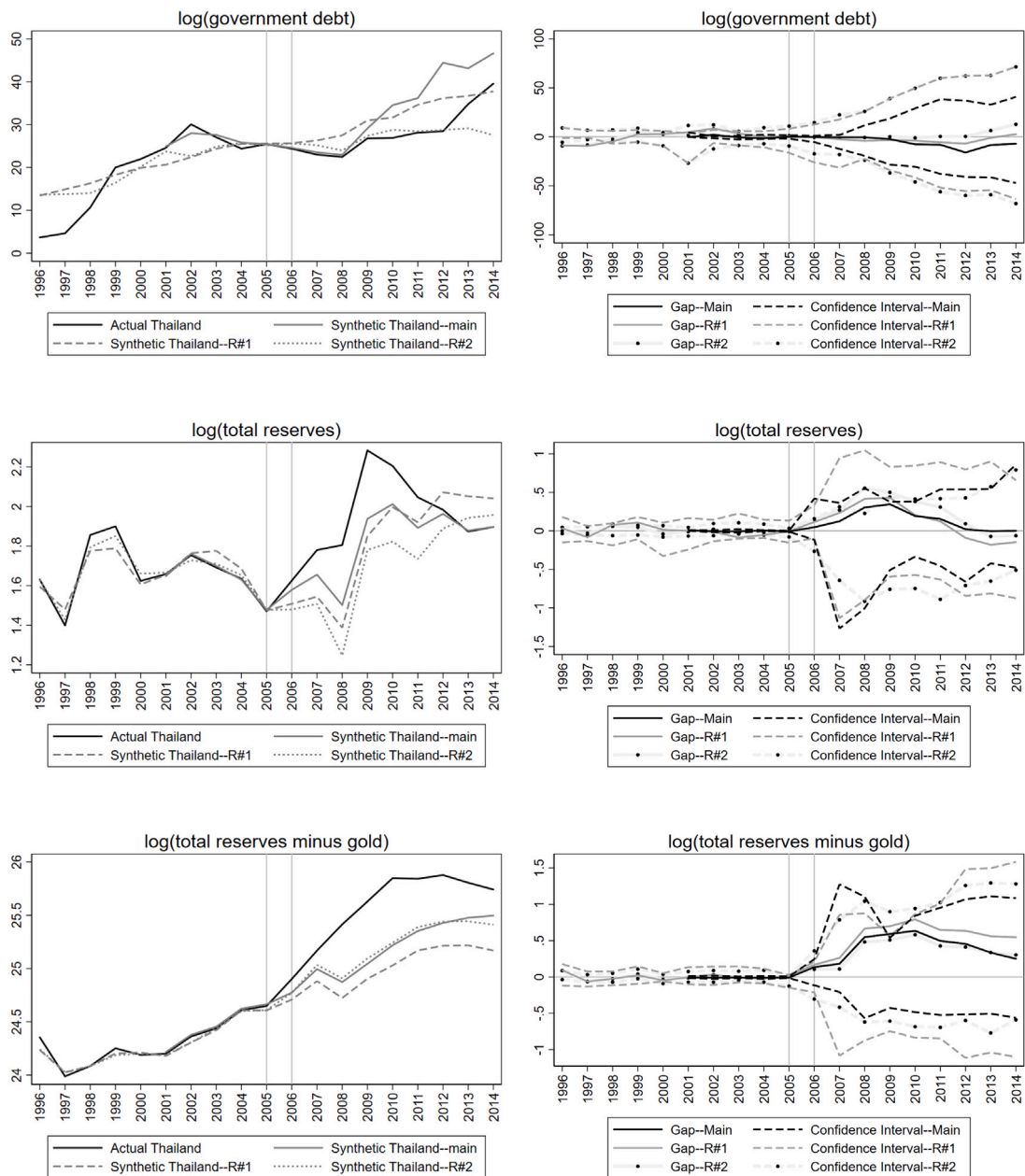


Fig. 7. Financial stability.

Note: The main analysis has 26 countries from 2001 to 2014. The additional analysis R#1 has 26 countries from 1996 to 2014. The additional analysis R#2 has 75 countries from 1996 to 2014.

5.3. Results on international responses

What reactions did Thailand's 2006 coup induce in other countries? I consider three variables: the number of tourists, the income from tourism, and the net bilateral aid flow from the United States.

The results on international tourism are shown in Fig. 9. Tourism in synthetic Thailand effectively replicates the trend of tourism in actual Thailand. Synthetic Thailand's tourism industry grows at a steady level, with a small increase in 2008. However, actual Thailand has a large increase in the number of international tourists and income from international tourism. Between 2005 and 2006, the number of international tourists increases by 17.8 percent, and income from international tourism increases by 31.4 percent.

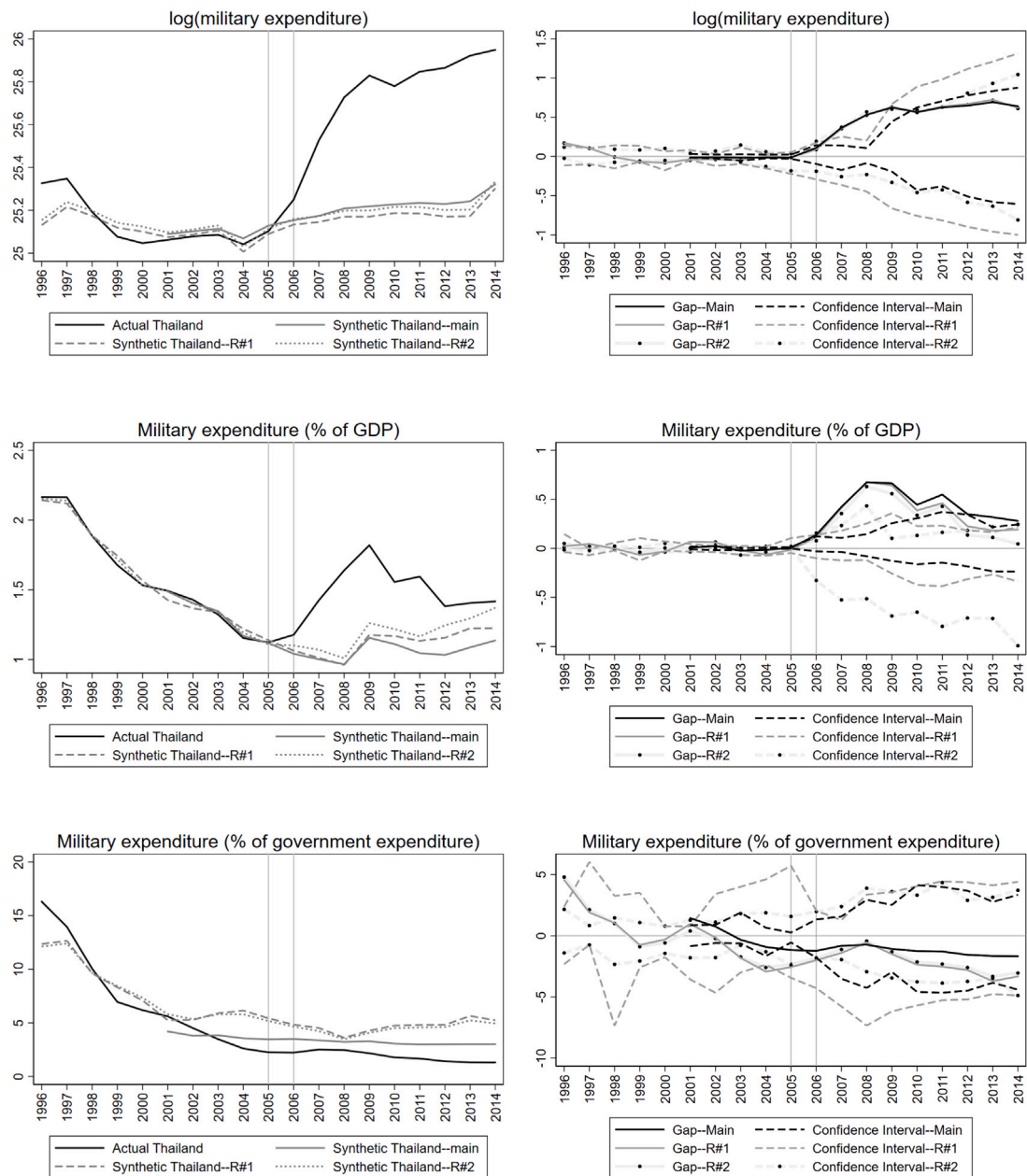


Fig. 8. Military Spending.

Note: The main analysis has 26 countries from 2001 to 2014. The additional analysis R#1 has 26 countries from 1996 to 2014. The additional analysis R#2 has 75 countries from 1996 to 2014.

The results of the placebo tests indicate that all synthetic Thailand suggest a statistically significant temporary positive effect of the coup on the number of international tourists in 2006. Synthetic Thailand from the analysis R#1 and R#2 detect statistically significant long-term positive effects on the number of international tourists between 2010 and 2014. However, synthetic Thailand from the main analysis suggests that the long-term positive effects of the coup on the number of international tourists are statistically significant in 2013 only.

Similarly, synthetic Thailand from the main analysis suggests that there is a statistically significant impact of the coup on the income from international tourism in the years 2006, 2007, and 2010 to 2014.

The statistically significant short-term effect of the coup on the number of international tourists is expected to be negative due to the temporary travel warnings announced by other countries. However, one explanation is that the military coup eliminated the

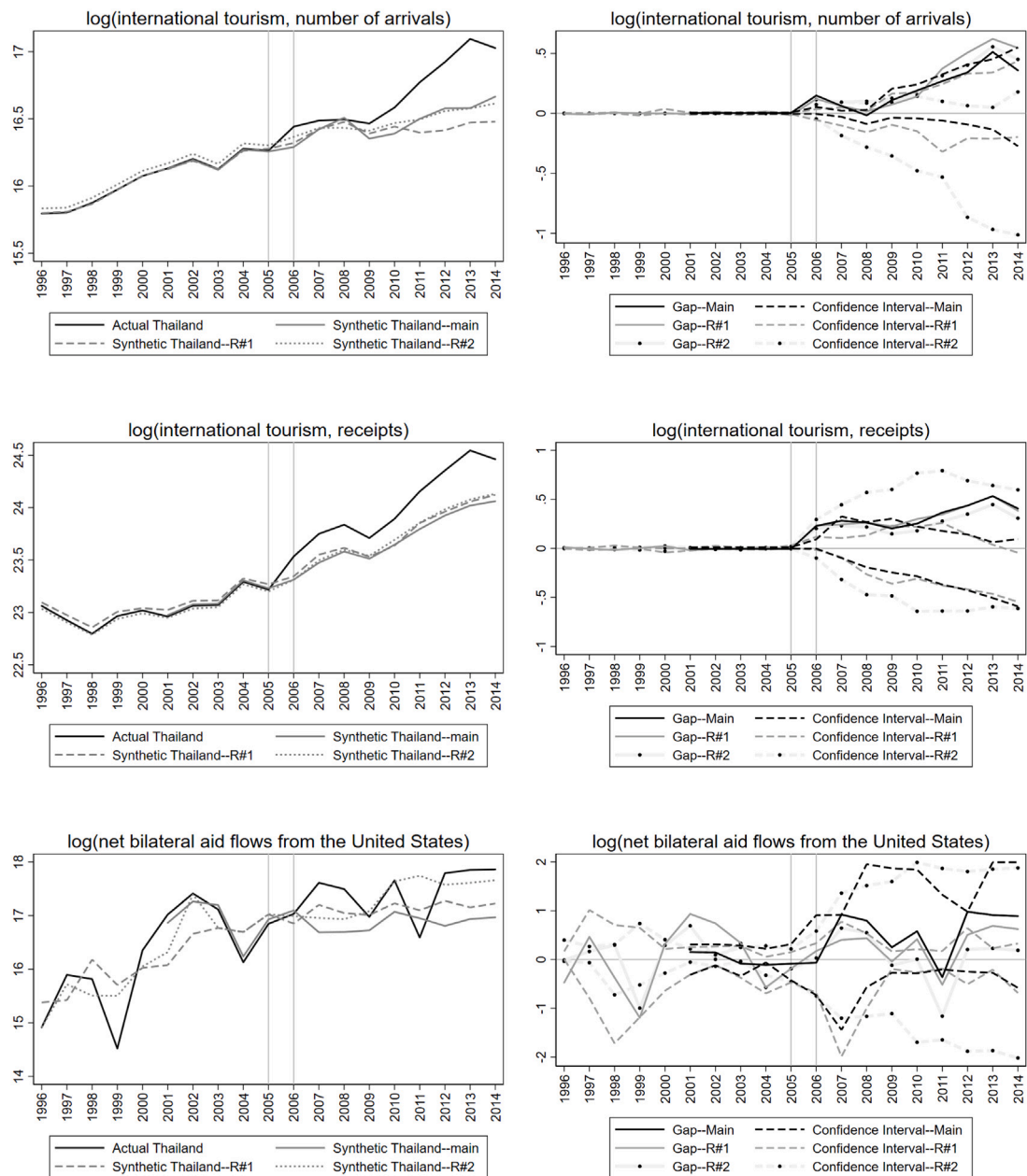


Fig. 9. International tourism and bilateral aid flows from the United States.

Note: The main analysis has 26 countries from 2001 to 2014. The additional analysis R#1 has 26 countries from 1996 to 2014. The additional analysis R#2 has 75 countries from 1996 to 2014.

political turmoil, allowing foreign visitors to travel to Bangkok without danger. In this case, the long-term effect would be that the number of travelers increases to the initial pre-coup level.

Next, I study the effect of Thailand's 2006 coup on net bilateral aid flows from the United States. Synthetic Thailand receives relatively constant bilateral aid flows from the United States. In reality, actual Thailand's aid flows from the United States increases between 2007 and 2008 and fluctuates between 2009 and 2012. The analysis does not find a statistically significant effect of the coup on bilateral aid flows from the United States.

The insignificant result can be due to the United States' attempt to maintain its relationship with Southeast Asia. As discussed in Section 3, the United States was put in a difficult position, having to decide whether or not to sanction the Thai military government in order to retain its position in the region.

6. Conclusion

Thailand has experienced political polarization since Thaksin Shinawatra became Thailand's prime minister. The confrontation between the "Yellow Shirts" and "Red Shirts" caused country-wide political tensions that ended with the military coup in September 2006, when Thaksin Shinawatra was in New York for the meeting of the UN General Assembly.

The paper investigates the effects of the coup on dimensions of Thailand's economy: (i) GDP and inflation, (ii) the components of GDP, (iii) labor market, (iv) financial stability, (v) military expenditure, and (vi) international tourism and bilateral aid flows from the United States. To predict how Thailand would have been if the 2006 coup had not occurred, I use the synthetic control method to construct synthetic Thailand from a linear combination of the countries in the control group.

This paper finds that the 2006 coup had positive and negative consequences. The 2006 military coup decreased consumption as a percentage of GDP but increased military expenditure and international tourism. Actual Thailand's military expenditure in 2009–2014 was 89.1–102.6 percent more than it would have been had the military coup not occurred. The magnitude is equal to 0.3–0.7 percent of GDP. According to the synthetic control method, the coup had no statistically significant effects on exports, imports, labor force participation, unemployment, government debts, and total reserves.

This paper sheds light on possible future research into the issue of whether military coups in the twenty-first century have different effects than they did in the past.

Declaration of competing interest

The author declares that there is no conflict of interest regarding the publication of this article.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.ejpoleco.2023.102475>.

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